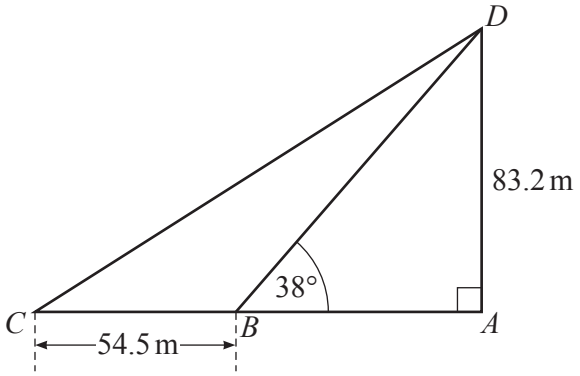
0580_w23_qp_41_Q5

5 **(a)**



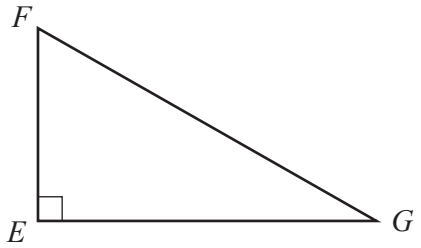
NOT TO SCALE

ACD is a right-angled triangle.
B is on *AC* and *BC* = 54.5 m.
AD = 83.2 m and angle *ABD* = 38°.

Calculate angle *ACD*.

Angle *ACD* = ... [5]

(b)



EFG is a right-angled triangle.
A circle can be drawn that passes through the three vertices of the triangle.

On the diagram, mark the position of the centre of the circle with a cross.
Explain how you decide.

.....

..... [2]

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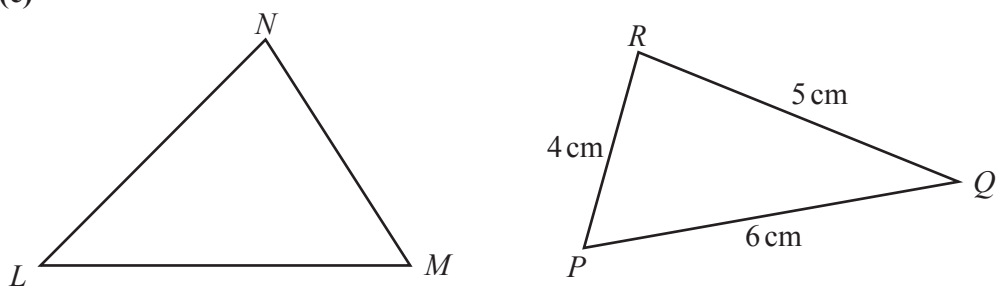
(c)



NOT TO SCALE

In triangle *LMN*, the ratio angle *L* : angle *M* : angle *N* = 4 : 5 : 6.
In triangle *PQR*, $PQ = 6\,\text{cm}$, $PR = 4\,\text{cm}$ and $QR = 5\,\text{cm}$.

Calculate the difference between the largest angle in triangle *PQR* and the largest angle in triangle *LMN*.

..... [7]